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Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
interface DeliveryCharge{
    double calculateDeliveryCharge(double foodCost);
}
interface CouponDiscount{
    double calculateDiscount(double foodCost,String couponCode);
}
class FoodOrder implements DeliveryCharge,CouponDiscount{
    public double calculateDeliveryCharge(double foodCost){
        if(foodCost>=500)
            return 0;
        else
            return 50;
    }
    public double calculateDiscount(double foodCost,String couponCode){
        if(couponCode.equals("SAVE10"))
            return foodCost*0.10;
        else
            return 0;
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        double foodCost = sc.nextDouble();
        sc.nextLine();
        String couponCode = sc.nextLine();

        FoodOrder f = new FoodOrder();

        double deliveryCharge = f.calculateDeliveryCharge(foodCost);
        double discount = f.calculateDiscount(foodCost, couponCode);
        double finalBill = foodCost + deliveryCharge - discount;
        System.out.println("Food Cost : " + foodCost);
        System.out.println("Delivery Charge : " + deliveryCharge);
        System.out.println("Discount : " + discount);
        System.out.println("Final Bill : " + finalBill);
        sc.close();
    }
}
```

INPUT

```
600
SAVE10
```

OUTPUT

```
Food Cost : 600.0
Delivery Charge : 0.0
Discount : 60.0
Final Bill : 540.0
```

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Room{
    int roomNumber;
    String guestName;
    double roomRent;
    Room(int r, String g, double rent){
        roomNumber=r;
        guestName=g;
        roomRent=rent;

    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int roomNumber = sc.nextInt();
        sc.nextLine();
        String guestName = sc.nextLine();
        double roomRent = sc.nextDouble();
        HotelBill h = new HotelBill(roomNumber, guestName, roomRent);
        h.display();
        sc.close();
    }
}
interface serviceCharge{
    double calculateServiceCharge();
}
class HotelBill extends Room implements serviceCharge{
    HotelBill(int r, String g, double rent){
        super(r, g, rent);
    }
    public double calculateServiceCharge(){
        if (roomRent >= 5000)
            return roomRent * 0.12;
        else
            return roomRent * 0.08;
    }
    void display(){
        double serviceCharge = calculateServiceCharge();
        double finalBill = roomRent + serviceCharge;

        System.out.println("Room Number : " + roomNumber);
        System.out.println("Guest Name : " + guestName);
        System.out.println("Room Rent : " + roomRent);
        System.out.println("Service Charge : " + serviceCharge);
        System.out.println("Final Bill : " + finalBill);
    }
}
```

INPUT

```
205
Ananya
6000
```

OUTPUT

```
Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0
```